Taesoo Kim

Georgia Institute of Technology, Atlanta, GA
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Research Interests

Advanced Semiconductor Packaging & Heterogeneous Integration

Designing chiplet-based, high-bandwidth systems for AI and high-performance computing, centered on **HBM-GPU interconnects** in **glass-core and interposer** architectures. My work integrates **signal/power-integrity (SI/PI) co-design** and **energy-per-bit optimization**, using full-wave EM and circuit simulation (Ansys HFSS/SIwave, Keysight ADS) together with **machine-learning-driven design methods**, to enable scalable, energy-efficient heterogeneous integration.

Education

Georgia Institute of Technology | Atlanta, GA
- Ph.D. in Electrical and Computer Engineering | Expected Graduation: Aug 2030
  - Area of research: SI/PI, Glass Packaging, Heterogeneous Integration
  - Advisor: Prof. Muhannad S. Bakir
  - GPA : 4.0 / 4.0

Korea Advanced Institute of Science and Technology (KAIST) | Daejeon, Korea
- M.S. in Electrical Engineering | Aug 2025
  - Area of research: SI/PI, High-Performance AI Hardware Design (HBM, Chiplets)
  - Advisor: Prof. Joungho Kim
  - GPA : 3.80 / 4.0

Korea Advanced Institute of Science and Technology (KAIST) | Daejeon, Korea
- B.S. in Electrical Engineering | Feb 2023
  - Magna Cum Laude
  - Selected for the Young Engineer Honor Society (YEHS), National Academy of Engineering of Korea (NAEK) by recommendation from the Dean of the College of Engineering.

Professional Experience

Cisco Systems, Inc. | San Jose, CA
- Hardware Engineer, PhD Intern (Signal Integrity Team) | Summer 2026
  - **High-Speed PCIe Signal Integrity:** Performed signal integrity analysis and lab-based measurements on PCIe interconnect boards to characterize and validate high-speed channel performance.
  - **Fiber Weave Effect Modeling:** Developed Python/MATLAB-based tools to quantify fiber-weave-induced skew and impedance variation on PCB transmission lines, correlated with full-wave EM (Ansys HFSS) and Cadence Allegro layout data.

Samsung Semiconductor, Inc. | San Jose, CA
- Intern, SI/PI Engineer | Summer 2024
  - **UCIe-A Channel Design:** Performed SI/PI analysis and optimized UCIe-A channels for high-speed chiplet communication, contributing to a successful **chip tape-out.**
  - **Link Performance Optimization:** Conducted full-wave EM simulations to minimize crosstalk and insertion loss, ensuring robust eye-margin for next-gen interconnects.

KAIST Terabyte Interconnection and Package Laboratory | Daejeon, Korea
- Graduated Research Assistant | 2023 - 2025
  - **Next-Gen HBM Architecture Design:** Spearheaded the design of advanced HBM architectures, including Twin Tower HBM (Best Paper Award, IEEE EDAPS 2024) and Quad Tower HBM (QT-HBM) using Glass Substrates (Best Poster Award, IEEE EPEPS 2025).
  - **Glass Substrate Optimization:** Achieved significant improvements in signal density and thermal management by implementing glass-based high-density interconnects.

- **Project Management:** Managed two industry-funded research projects focused on SI/PI modeling for 2.5D/3D heterogenous integration.

KAIST Computer, Architecture, System and Technology Lab | Daejeon, Korea

- Research Assistant | 2022
  - DMA Controller Design: Developed high-efficiency DMA architectures, optimizing data transfer between SRAM and logic units for AI accelerators.

Publications

- **Taesoo Kim**, et al., "A Design-Space Exploration of Active Interposer Architectures for Long-Reach Signaling in Large-Area Advanced Packages," *IEEE Electronic Components and Technology Conference (ECTC)* (2026)
- **Taesoo Kim**, et al., "Quad Tower High Bandwidth Memory (QT-HBM) with Glass Substrate and Local Silicon Bridge," *IEEE Conference on Electrical Performance of Electronic Packaging and Systems (EPEPS)* (2025), **Best Poster Award**
- **Taesoo Kim**, et al., "Design and Verification of a UCIe-A Mimic Channel for Early-Stage Development of UCIe-Based Systems," *IEEE Electronic Components and Technology Conference (ECTC)* (2025)
- **Taesoo Kim**, et al., "Design and Analysis of Twin Tower High Bandwidth Memory (HBM) for Large Memory Capacity and High Bandwidth System," *IEEE Electrical Design of Advanced Packaging and Systems* (2024), **Best Paper Award**
- Hyunjun An, Taesoo Kim (9th author), et al., "Multi-Chiplet Power Delivery Network Co-Optimization Considering Current Spectrum by Deep Reinforcement Learning-Based Decoupling Capacitor Placement," *IEEE Electronic Components and Technology Conference* (2025)
- Haeseok Suh, Taesoo Kim (9th author), et al., "Generative Model Based Multi-Layer PDN Impedance Estimation With Multi-Power Domain," *IEEE Electronic Components and Technology Conference* (2025)
- Hyunah Park, Taesoo Kim (9th author), et al., "Design and Analysis of Extended Scale Cache (ESC) Stacked-GPU-HBM Module Architecture Considering Power Integrity (PI)," *IEEE EPEPS* (2024)
- Hyunjun An, Taesoo Kim (8th author), et al., "Eye-Diagram Edge Estimation (EEE) Network for Through Silicon Via Design in Next-Generation High Bandwidth Memory," *IEEE EPEPS* (2024)
- Haeseok Suh, Taesoo Kim (9th author), et al., "Design and Analysis of L3 Cache Embedded-GPU-High Bandwidth Memory Architecture with Reduced Energy and Latency for AI Computing," *IEEE EPEPS* (2024)

Honors & Awards

- Selected, SK hynix Global Ph.D. Fellowship, SK hynix Inc., Korea (2026)
- Best Poster Award, IEEE Conference on Electrical Performance of Electronic Packaging and Systems (EPEPS) (2025)
- Best Paper Award, IEEE Electrical Design of Advanced Packaging and Systems (EDAPS) (2024)
- National Science & Technology Scholarship, Republic of Korea

Technical Skills

- **Simulation/EM Tools:** Ansys HFSS, SIwave, Maxwell, Q3D, Keysight ADS, Cadence Allegro, Cadence Virtuoso
- **Languages/Frameworks:** Python(NumPy, SciPy, Scikit-Learn), Verilog, Matlab
- **Machine Learning and AI Techniques:** Reinforcement Learning for optimization

Leadership & Activities

Leadership Roles

- President, Graduate Student Association of Electrical Engineering, KAIST (2024)
- President, Young Engineers Honor Society (YEHS), National Academy of Engineering of Korea, Seoul, Korea (2022)
- Vice President, Korean Student Association, Georgia Tech, Atlanta, GA (2025)

Military Service

- Detection Analysis Specialist (Sergeant), ROK Defense Ground Operations Command, Korea (2020–2021)